



PENRITH HIGH SCHOOL

**2015
HSC TRIAL EXAMINATION**

Mathematics Extension 2

General Instructions:

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In questions 11 – 16, show relevant mathematical reasoning and/or calculations
- Answer all Questions on the writing sheets provided

Total marks–100

SECTION I

Pages 3–7

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

SECTION II

Pages 9–15

90 marks

- Attempt Questions 11–16
- Allow about 2 hours 45 minutes for this section

Student Name: _____

Teacher Name: _____

This paper MUST NOT be removed from the examination room

Assessor: Mr Ferguson

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1 Which pair of coordinates gives the foci of $4x^2 - 25y^2 = 100$?

(A) $(\pm\sqrt{29}, 0)$

(B) $\left(\pm\frac{\sqrt{29}}{5}, 0\right)$

(C) $\left(\pm\frac{\sqrt{21}}{5}, 0\right)$

(D) $(\pm\sqrt{21}, 0)$

2 What are the values of a and b for which the following identity is true?

$$\frac{3x^2 + 7}{(x^2 + 9)(x^2 + 4)} = \frac{a}{x^2 + 9} + \frac{b}{x^2 + 4}$$

(A) $a = 1$ and $b = 2$

(B) $a = 4$ and $b = -1$

(C) $a = 1$ and $b = -2$

(D) $a = 4$ and $b = 1$

3 The region in the first quadrant between the x -axis and $y = 6x - x^2$ is rotated about the y -axis. The volume of this solid of revolution is.

(A) $\pi \int_0^6 (6x - x^2) dx$

(B) $\pi \int_0^6 x(6x - x^2)^2 dx$

(C) $2\pi \int_0^6 x(6x - x^2) dx$

(D) $\pi \int_0^6 (3 + \sqrt{9 - y})^2 dy$

4 Which expression is equal to $\int \frac{dx}{\sqrt{4x-x^2}}$?

(A) $\ln\left[(x-2)+\sqrt{6-x}\right]+c$

(B) $\ln\left[(x-2)+\sqrt{6+x}\right]+c$

(C) $\sin^{-1}\frac{x-2}{2}+c$

(D) $\cos^{-1}\frac{x-2}{2}+c$

5 The polynomial $4x^3+x^2-3x+5=0$ has roots α, β and γ . Which polynomial equation has roots $\frac{\alpha}{2}, \frac{\beta}{2}$ and $\frac{\gamma}{2}$?

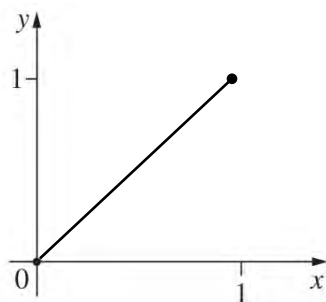
(A) $8x^3+2x^2-6x+10=0$

(B) $2x^3+x^2-6x+5=0$

(C) $32x^3+4x^2-6x+5=0$

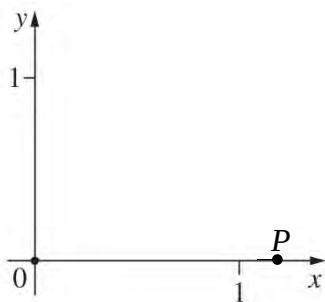
(D) $5x^3-3x^2+x+4=0$

- 6 The Argand diagram below shows the complex number z .

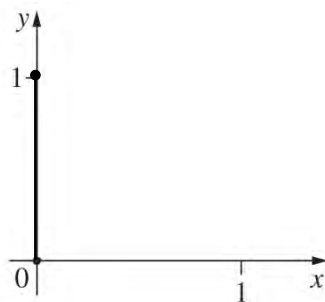


Which diagram best represents the locus of P such that $P = |z|$?

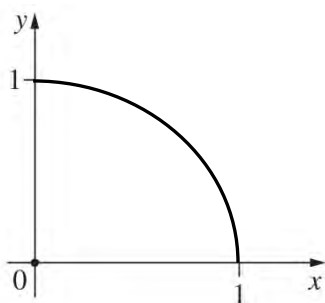
(A)



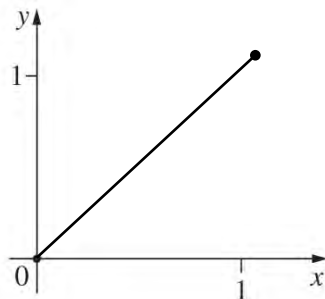
(B)



(C)

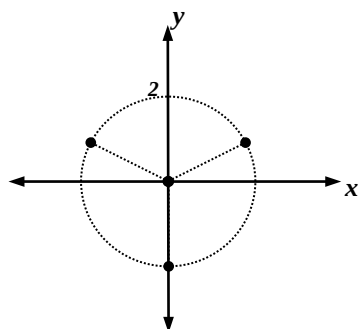


(D)

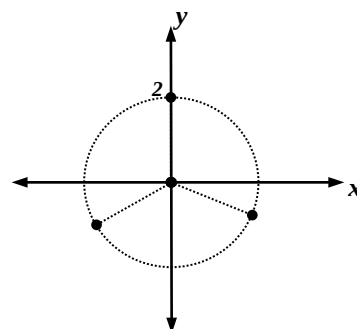


- 7 Which diagram best represent the cube roots of $8i$?

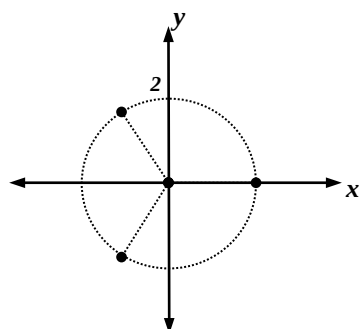
(A)



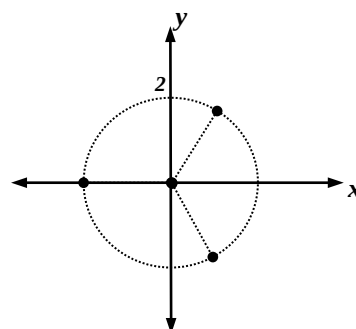
(B)



(C)

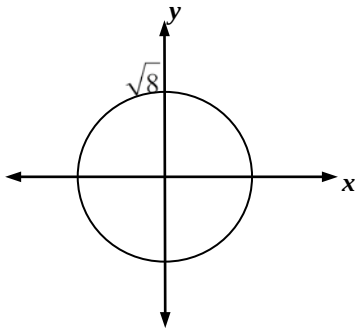


(D)

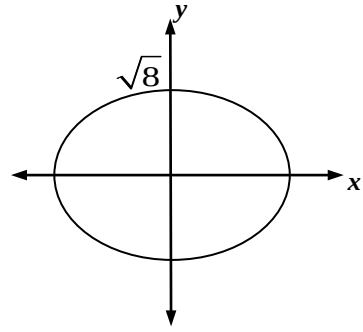


8 Which diagram best represents $z^2 + \bar{z}^2 = 16$

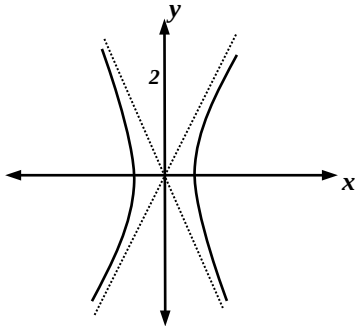
(A)



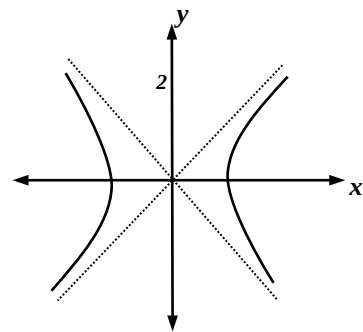
(B)



(C)



(D)



9 A particle of mass m falls from rest under gravity and the resistance to its motion is mkv^2 , where v is its speed and k is a positive constant. Which of the following is the correct expression for square of the velocity where x is the distance fallen?

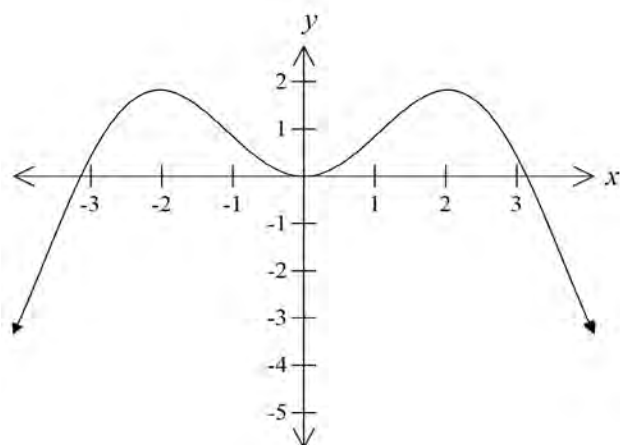
(A) $v^2 = \frac{g}{k}(1 - e^{-2kx})$

(B) $v^2 = \frac{g}{k}(1 + e^{-2kx})$

(C) $v^2 = \frac{g}{k}(1 - e^{2kx})$

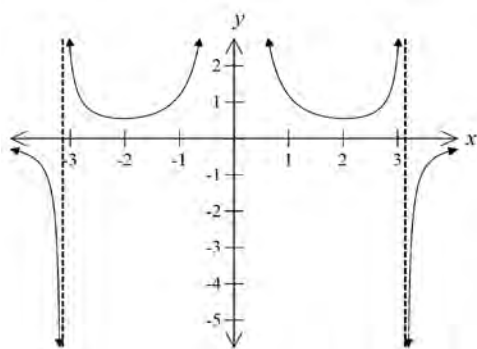
(D) $v^2 = \frac{g}{k}(1 + e^{2kx})$

10 The diagram shows the graph of the function $y = f(x)$.

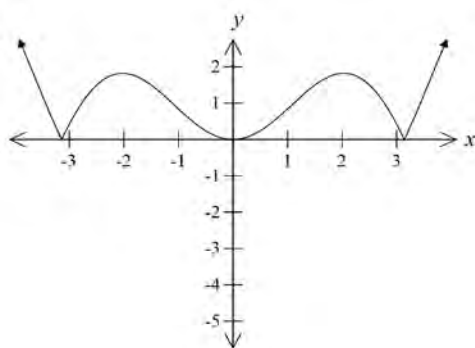


Which of the following is the graph of $y = |f(x)|$?

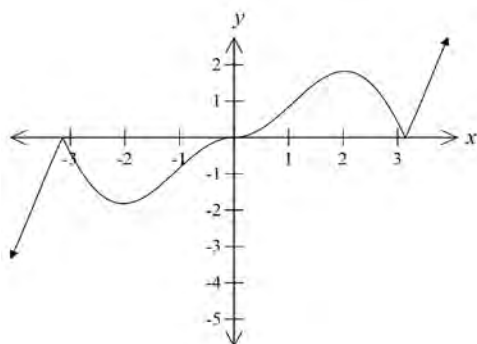
(A)



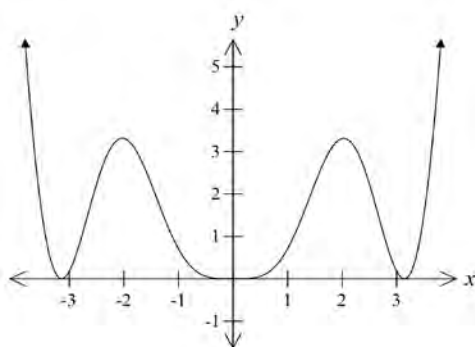
(B)



(C)



(D)



Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question on a new writing sheet. Extra writing sheets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing sheet.

a) Given that $z = \sqrt{3} + \frac{1+i}{1-i}$ find:

(i) $\operatorname{Im}(z)$ **1**

(ii) $\bar{z}$ **1**

(iii) z in modulus argument form **2**

b) Sketch separately the following loci in an Argand plane.

(i) $2|z - (1+i)| = |z - (4+i)|$ **2**

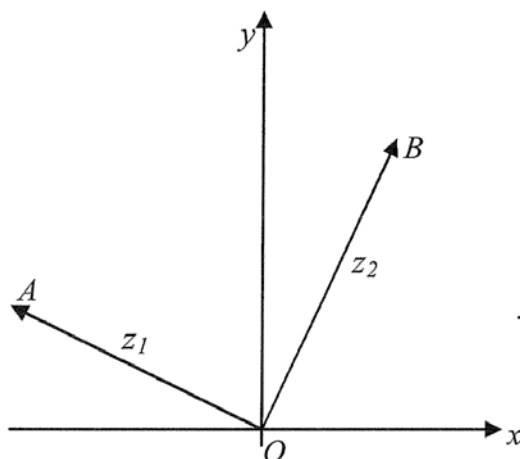
(ii) $\left\{ z : 0 \leq \arg(z + 4 + i) \leq \frac{2\pi}{3} \text{ and } |z + 4 + i| \leq 4 \right\}$ **3**

c) Find $\int_0^{\frac{\pi}{2}} \frac{dx}{1 + \tan x}$ **3**

d) For $x > 0, y > 0, z > 0$ show that $x + y + z + \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \geq 6$ **3**

Question 12 (15 marks) Use a SEPARATE writing sheet.

a)



In the Argand diagram $\overrightarrow{OA}$ and $\overrightarrow{OB}$ represent complex numbers $z_1 = 2\left(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\right)$ and $z_2 = 2\left(\cos\frac{5\pi}{12} + i\sin\frac{5\pi}{12}\right)$ respectively.

i) Show that $\triangle OAB$ is equilateral 3

ii) Explain why $z_2 - z_1$ is equal to z_2 rotated by $\frac{\pi}{3}$ radians 1

iii) Express $z_2 - z_1$ in modulus-argument form. 3

b) Solve $z^2 = -8 - 6i$ 2

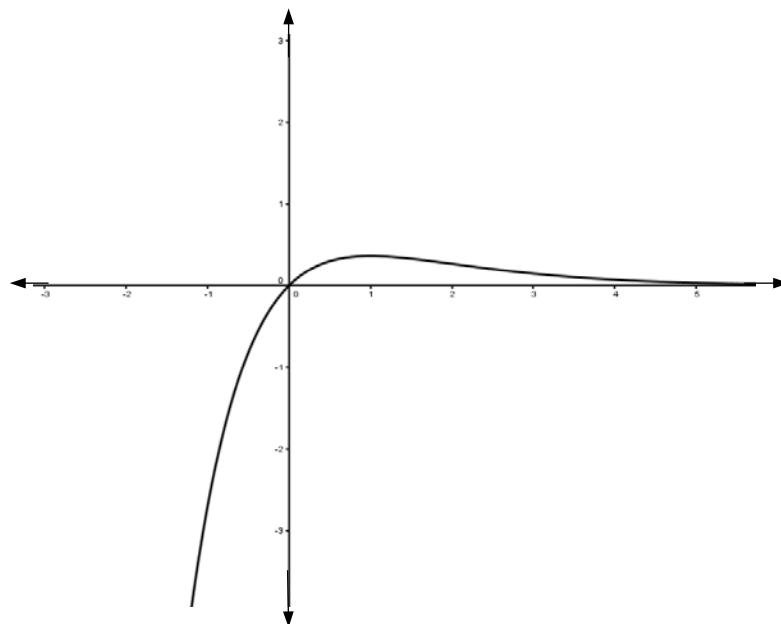
c) (i) Show that the recurrence (reduction) formula for $I_n = \int \tan^n x \, dx$ is 3

$$I_n = \frac{1}{n-1} \tan^{n-1} x - I_{n-2}.$$

(ii) Hence evaluate $\int_0^{\frac{\pi}{4}} \tan^3 x \, dx$ 3

Question 13 (15 marks) Use a SEPARATE writing sheet.

- a) The diagram shows the graph of $y = f(x)$. The graph has a horizontal asymptote at $y = 0$.



Draw separate one-third page sketches of the graphs of the following:

(i) $y = (f(x))^2$ 2

(ii) $y = \frac{1}{f(x)}$ 2

(iii) $y = x f(x)$ 2

(iv) $y = f(|x|)$ 2

- b) The ellipse, E , has equation $9x^2 + 25y^2 = 225$.

P is any point on the ellipse and A and B are the points $(5,0)$ and $(-5,0)$ respectively.
 AP , produced if necessary, meets the y axis in Q , and BP , also produced if necessary,
meets the y axis in R

The tangent at P meets the y axis in T

- (i) Find the eccentricity 1

- (ii) Sketch the ellipse, E , showing the coordinates of its foci. 2

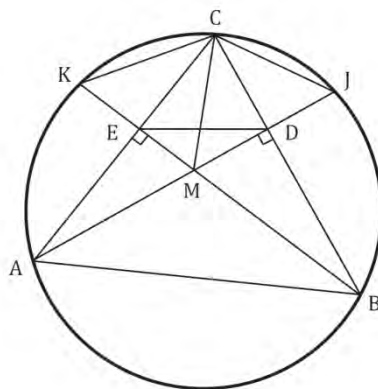
- (iii) Given that the equation of the tangent at P is $\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$.

Prove that T is the midpoint of QR 4

Question 14 (15 marks) Use a SEPARATE writing sheet.

- a) For any non-zero real number t , the point $\left(t, \frac{1}{t}\right)$ lies on the graph of $y = \frac{1}{x}$.
- (i) Show that $9xy = 1$ is the equation of the locus of the point that divides the straight line joining $\left(t, \frac{1}{t}\right)$ and $\left(-t, \frac{-1}{t}\right)$ in the ratio of 1:2 respectively, as t varies. 2
- (ii) Show that the equation of the tangent to $y = \frac{1}{x}$ at the point $\left(t, \frac{1}{t}\right)$ may be written in the form $t^2y - 2t + x = 0$ 2
- (iii) $R(0, h)$ is a point on the y axis. Show that there is exactly one point on the hyperbola $y = \frac{1}{x}$ with tangents that pass through R 2
- b) Find $\int \frac{dx}{a^2 \cos^2 x + b^2 \sin^2 x}$ 3

c)



In the diagram ABC is a triangle inscribed in a circle. The altitude AD is produced to meet the circle at J . The altitude BE is produced to meet the circle at K and the two altitudes intersect at M .

- i) Copy the diagram onto your answer sheet
- ii) Show that $ABDE$ and $CEMD$ are cyclic 2
- iii) Prove that AC bisects $\angle KCM$ 2
- iv) Prove that $KC = JC$ 2

Question 15 (15 marks) Use a SEPARATE writing sheet.

- a) A particle of mass m kg is set in motion, with speed $u \text{ ms}^{-1}$ and moves in a straight line before coming to rest. At time t seconds the particle has displacement x metres from its starting point O , velocity $v \text{ ms}^{-1}$ and acceleration $a \text{ ms}^{-2}$

The resultant force acting on the particle directly opposes its motion and has magnitude $m(1+v)$ Newtons.

(i) Show that $a = -(1+v)$ 1

(ii) Find expressions for

1. x in terms of v 2

2. v in terms of t 2

3. x in terms of t 2

(iii) Show that $x + v + t = u$ 2

(iv) Find the distance travelled and time taken by the particle in coming to rest. 2

- b) Given that $z = 1 - 2i$ is a factor of the equation $P(z) = z^4 - z^3 + 6z^2 - z + 15$

(i) Factorise $P(z)$ into real quadratic factors 2

(ii) Solve for $P(z) = 0$ for z 2

Question 16 (15 marks) Use a SEPARATE writing sheet.

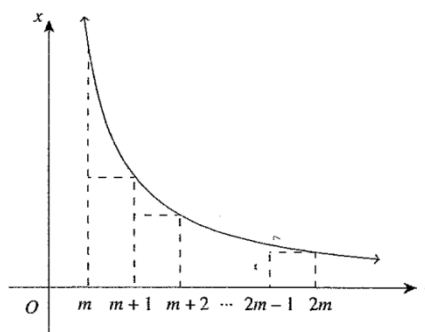
a) (i) Prove that $\frac{1}{2p+1} + \frac{1}{2p+2} > \frac{1}{p+1}$, for all $p > 0$ 2

(ii) Consider the statement

$$\psi(m): \frac{1}{m+1} + \frac{1}{m+2} + \dots + \frac{1}{2m} \geq \frac{37}{60}$$
3

Show that by mathematical induction that $\psi(m)$ is true for all integers $m \geq 3$.

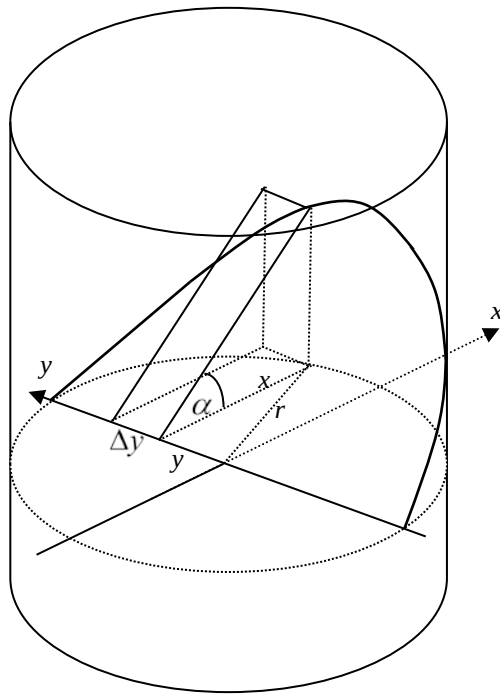
(iii) The diagram below shows the graph of $x = \frac{1}{t}$, for $t > 0$



(iv) By comparing areas, show that $\int_m^{m+1} \frac{1}{t} dt > \frac{1}{m+1}$ 2

(v) Hence, without using a calculator, show that $\log_e 2 > \frac{37}{60}$ 3

- b) A wedge is cut from a right circular cylinder of radius r by two planes, one perpendicular to the axis of the cylinder while the second makes an angle α with the first and intersects it at the centre of the cylinder.



A is the area of the triangle that forms one face of the slice.

i) Show that $A = \frac{1}{2}(r^2 - y^2) \tan \alpha$. 2

ii) Hence show that the volume of the wedge is $\frac{2}{3}r^3 \tan \alpha$ 3

End of Exam

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a \neq 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE: $\ln x = \log_e x, x > 0$

Student Number: _____

Teacher Name: _____

Section I - Multiple Choice

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
 correct

**Start
Here** →

1. A ☐ B ☐ C ☐ D ☐
2. A ☐ B ☐ C ☐ D ☐
3. A ☐ B ☐ C ☐ D ☐
4. A ☐ B ☐ C ☐ D ☐
5. A ☐ B ☐ C ☐ D ☐
6. A ☐ B ☐ C ☐ D ☐
7. A ☐ B ☐ C ☐ D ☐
8. A ☐ B ☐ C ☐ D ☐
9. A ☐ B ☐ C ☐ D ☐
10. A ☐ B ☐ C ☐ D ☐

Multiple Choice

1) $\frac{x^2}{25} - \frac{y^2}{4} = 1$

$a^2 = 25 \quad b^2 = 4$

foci $(\pm ae, 0)$

$b^2 = a^2(e^2 - 1)$

$4 = 25(e^2 - 1)$

$e^2 - 1 = \frac{4}{25}$

$e^2 = \frac{29}{25}$

$e = \pm \frac{\sqrt{29}}{5}$

$(ae, 0) = (\pm \sqrt{29}, 0)$

= (A)

2) $a(x^2 + 4) + b(x^2 + 9) = 3x^2 + 7$

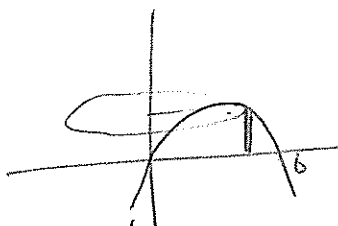
when $a = 4 \quad b = -1$

$4x^2 + 16 - x^2 - 9$

$= 3x^2 + 7$

∴ (B)

3)



$2\pi \int_0^6 xy \, dx$

$2\pi \int_0^6 x(6x - x^2) \, dx$

(C)

4) $\int \frac{dx}{\sqrt{4 - (x-2)^2}}$

$= \sin^{-1} \frac{x-2}{2} + C$ (standard integral)

= (C)

5) $4(2x)^3 + (2x)^2 - 3(2x) + 5 = 0$

$32x^3 + 4x^2 - 6x + 5 = 0$

(C)

6) (A) $p = |1+i|$

$p = \sqrt{2}$

7) A

8) $(x+iy)^2 + (x-iy)^2 = 16$

$x^2 + 2xyi - y^2 + x^2 - 2xyi - y^2 = 16$

$2x^2 - 2y^2 = 16$

$x^2 - y^2 = 8$ (rectangular hyperbola)

D

9) (A) $m\ddot{x} = mg - kv^2$

$\ddot{x} = g - kv^2$

$v \frac{dv}{dx} = g - kv^2$

$\frac{dv}{dx} = \frac{g}{v} - kv$

$\frac{dx}{dv} = \frac{v}{g - kv^2}$

$x = -\frac{1}{2k} \ln(g - kv^2) + C$ when $x=0 \quad v=0$

$x = -\frac{1}{2k} \ln(g - kv^2)$

$\therefore v^2 = \frac{g}{k} (1 - e^{-2kx})$

(10) (B)

Question 11

$$a) \quad z = \sqrt{3} + \frac{1+i}{1-i}$$

$$\frac{1+i}{1-i} = \frac{1+i}{1-i} \times \frac{1+i}{1+i}$$

$$= \frac{1+i^2+2i}{1-i^2}$$

$$= \frac{2i}{2}$$

$$= i$$

$$\therefore z = \sqrt{3} + i$$

$$(i) \quad \operatorname{Im}(z) = 1$$

$$(ii) \quad \bar{z} = \sqrt{3} - i$$

$$(iii) \quad z = \sqrt{3} + i$$

$$r^2 = (\sqrt{3})^2 + 1^2$$

$$= 4$$

$$\therefore r = 2$$

$$z = 2\left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)$$

$$= 2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$$

$$= 2 \operatorname{cis} \frac{\pi}{6}$$

$$b) (i) 2|z-(1+i)| = |z-(4+i)|.$$

squaring both sides

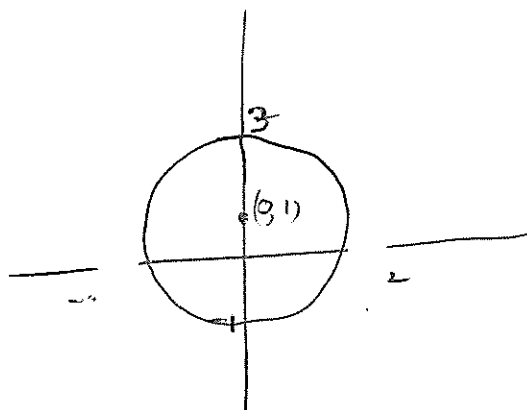
$$4[(x-1)^2 + (y-1)^2] = (x-4)^2 + (y-1)^2$$

$$4(x^2 - 2x + 1 + y^2 - 2y + 1) = (x^2 - 8x + 16 + y^2 - 2y + 1) = 0$$

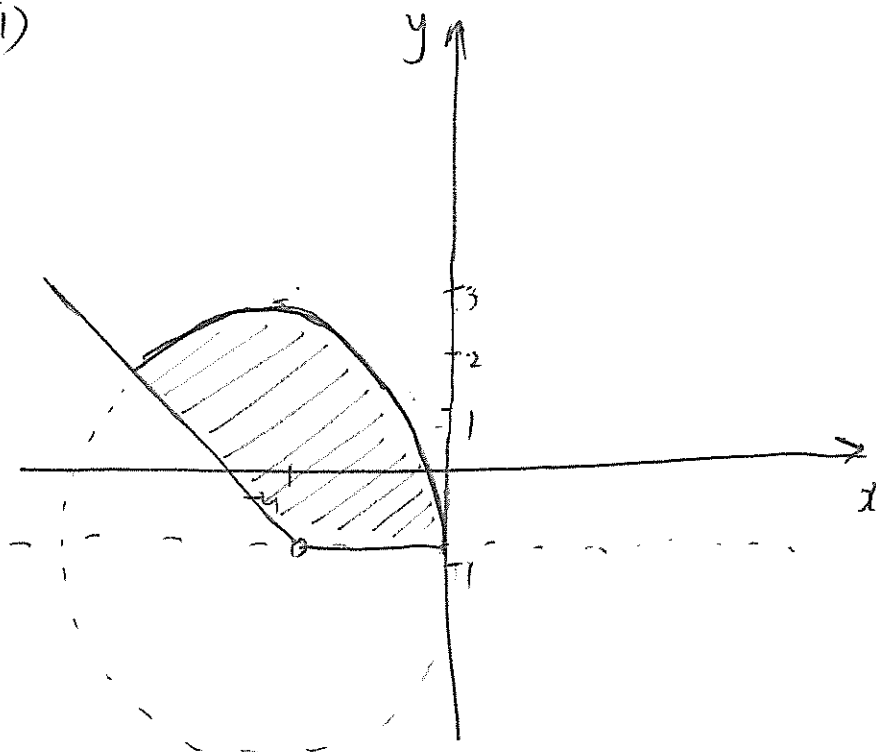
$$3x^2 - 9 + 3y^2 - 6y = 0$$

$$x^2 + (y-1)^2 = 4$$

$\therefore$ circle centre $(0, 1)$ radius 2.



(ii)



c)

$$I = \int_0^{\frac{\pi}{2}} \frac{dx}{1+\tan x}$$

$$= \int_0^{\frac{\pi}{2}} \frac{dx}{1+\tan(\frac{\pi}{2}-x)}$$

$$\int_a^b f(x) dx$$

$$= \int_a^b f(a-x) dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{dx}{1+\cot x}$$

$$= \int_0^{\frac{\pi}{2}} \frac{\tan x}{1+\tan x} dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{1+\tan x - 1}{1+\tan x} dx$$

$$= \int_0^{\frac{\pi}{2}} (1 - I) dx$$

$$2I = \int_0^{\frac{\pi}{2}} 1 dx$$

$$2I = \frac{\pi}{2}$$

$$I = \frac{\pi}{4}$$

alternative solution

d) For $x > 0, y > 0, z > 0$

$$x + y + z + \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \geq 6$$

$$(a-1)^2 \geq 0$$

$$a^2 - 2a + 1 \geq 0$$

$$a^2 + 1 \geq 2a$$

$$a + \frac{1}{a} \geq 2 \text{ as } a \neq 0$$

$$\therefore x + y + z + \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$$

$$= (x + \frac{1}{x}) + (y + \frac{1}{y}) + (z + \frac{1}{z})$$

$$\geq 2 + 2 + 2$$

$$\geq 6$$

$$I = \int_0^{\frac{\pi}{2}} \frac{dx}{1 + \frac{\sin x}{\cos x}}$$

$$= \int_0^{\frac{\pi}{2}} \frac{\cos x}{\cos x + \sin x} dx \quad \text{--- (1)}$$

$$= \int_0^{\frac{\pi}{2}} \frac{\cos(\frac{\pi}{2}-x)}{\cos(\frac{\pi}{2}-x) + \sin(\frac{\pi}{2}-x)} dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{\sin x}{\sin x + \cos x} dx$$

$$(1) + (2) = \int_0^{\frac{\pi}{2}} \frac{\sin x + \cos x}{\sin x + \cos x} dx$$

$$2I = \int_0^{\frac{\pi}{2}} 1 dx$$

$$2I = [x]_0^{\frac{\pi}{2}}$$

$$2I = \frac{\pi}{2}$$

$$I = \frac{\pi}{4}$$

Question 12

$$\begin{aligned} \text{(i)} \quad z_2 \times \text{cis} \frac{\pi}{3} &= 2 \text{cis} \frac{5\pi}{12} \text{cis} \frac{\pi}{3} \\ &= 2 \text{cis} \left(\frac{5\pi}{12} + \frac{\pi}{3} \right) \\ &= 2 \text{cis} \frac{3\pi}{4} \end{aligned}$$

$\therefore z_1$ is z_2 rotated
by $\frac{\pi}{3}$ clockwise

$$\therefore \angle AOB = \frac{\pi}{3}$$

$$|z_1| = 2$$

$$|z_2| = 2$$

$\therefore$ base angles $\angle OAB = \angle ABO$

since angle sum of triangle is π .

$\therefore \triangle OAB$ equilateral

Alternatively $|z_1| = |z_2| = 2$

show $|z_2 - z_1| = 2$.

$$\begin{aligned} |AB| &= |z_2 - z_1| \\ &= 2 \left| \text{cis} \frac{5\pi}{12} - \text{cis} \frac{3\pi}{4} \right| \end{aligned}$$

$$= 2 \sqrt{\left(\cos \frac{5\pi}{12} - \cos \frac{3\pi}{4} \right)^2 + \left(\sin \frac{5\pi}{12} - \sin \frac{3\pi}{4} \right)^2}$$

$$= 2 \sqrt{\cos^2 \frac{5\pi}{12} + \cos^2 \frac{3\pi}{4} - 2 \cos \frac{5\pi}{12} \cos \frac{3\pi}{4} + \sin^2 \frac{5\pi}{12} + \sin^2 \frac{3\pi}{4} - 2 \sin \frac{5\pi}{12} \sin \frac{3\pi}{4}}$$

$$= 2 \sqrt{1 + 1 - 2 \left(\cos \frac{5\pi}{12} \cos \frac{3\pi}{4} + \sin \frac{5\pi}{12} \sin \frac{3\pi}{4} \right)}$$

$$= 2 \sqrt{2 \left(1 - \cos \left(-\frac{\pi}{3} \right) \right)} \quad \text{since } \cos(-x) = \cos x$$

$$= 2 \sqrt{2 \left(1 - \frac{1}{2} \right)}$$

$$= 2$$

$\therefore \triangle OAB$ is equilateral

(ii) Since $\triangle OAB$ is an equilateral triangle

$$\therefore \angle AOB = \angle OBA = \angle BAO = \frac{\pi}{3}$$

$\therefore z_2 - z_1$ is obtained by rotating z_2 by $\frac{\pi}{3}$ radians

$$\begin{aligned} \text{(iii)} \quad z_2 - z_1 &= z_2 \operatorname{cis}\left(\frac{\pi}{3}\right) \\ &= 2 \operatorname{cis}\left(\frac{5\pi}{12}\right) \operatorname{cis}\left(\frac{\pi}{3}\right) \\ &= 2 \operatorname{cis}\frac{\pi}{12} \end{aligned}$$

b)

$$z^2 = -8 - 6i$$

$$(x+iy)^2 = -8-6i$$

$$x^2 + 2xyi - y^2 = -8 - 6i$$

$$x^2 - y^2 = -8$$

$$2xy = -6$$

$$y = -\frac{3}{x}$$

$$x^2 - \frac{9}{x^2} = -8$$

$$x^4 - 9 = -8x^2$$

$$x^4 + 8x^2 - 9 = 0$$

$$(x^2 - 1)(x^2 + 9) = 0$$

$$x = \pm 1 \quad \therefore y = \mp 3$$

$$z = (1-3i) \text{ or } z = (-1+3i)$$

$$C' (i) \quad I_n = \int \tan^n x dx$$

$$= \int \tan^{n-2} x \tan^2 x dx$$

$$= \int \tan^{n-2} x (\sec^2 x - 1) dx$$

$$= \int \tan^{n-2} x \sec^2 x dx - I_{n-2}$$

$$= \frac{\tan^{n-1} x}{n-1} - I_{n-2}$$

Reverse chain rule.

If unsure let do by substitution

$$A = \tan x$$

$$dA = \sec^2 x dx$$

$$\therefore \int A^{n-2} dA = \frac{A^{n-1}}{n-1}$$

$$= \frac{\tan^{n-1} x}{n-1}$$

$$(ii) \quad \int_0^{\frac{\pi}{4}} \tan^3 x dx$$

$$I_1 = \int_0^{\frac{\pi}{4}} \tan x dx$$

$$= \left[\ln(\cos x) \right]_0^{\frac{\pi}{4}}$$

$$= -\ln\left(\frac{1}{\sqrt{2}}\right)$$

$$I_3 = \frac{1}{3-1} \tan^2 x - I_{3-2}$$

$$= \frac{1}{2} \tan^2 x - I_1$$

$$= \frac{1}{2} \left[\tan^2 \frac{\pi}{4} - \tan^2 0 \right] + \ln\left(\frac{1}{\sqrt{2}}\right)$$

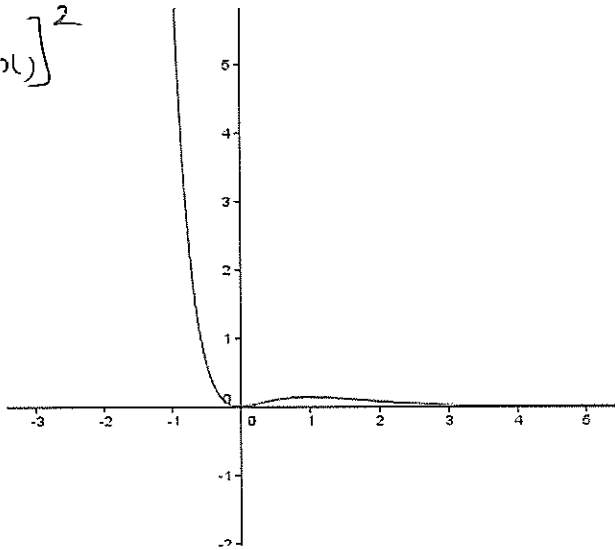
$$= \frac{1}{2} (1) + \ln \frac{1}{\sqrt{2}}$$

$$= \frac{1}{2} - \ln 2^{\frac{1}{2}}$$

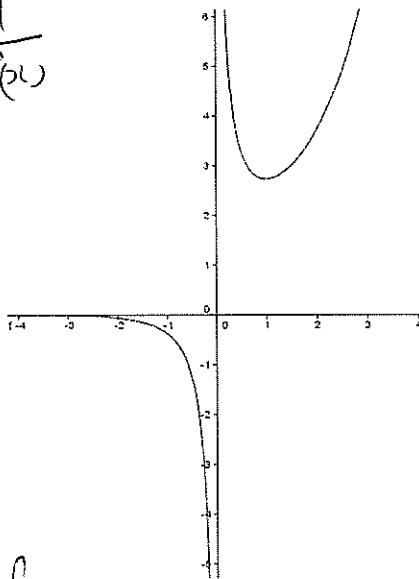
$$= \frac{1}{2} (1 - \ln 2)$$

Question 13

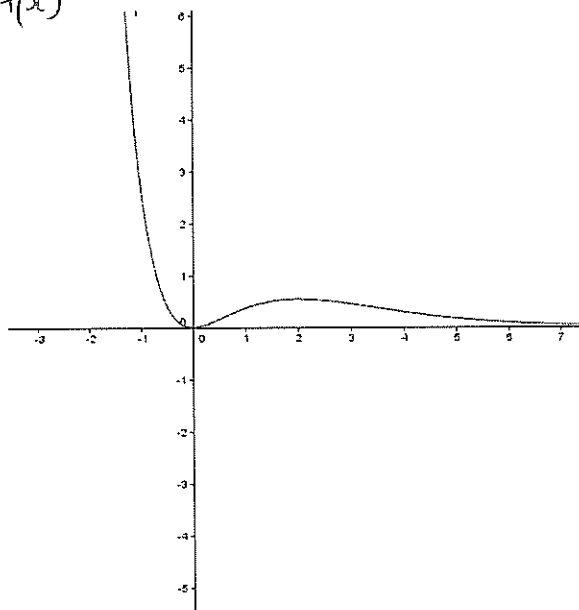
$$y = [f(x)]^2$$



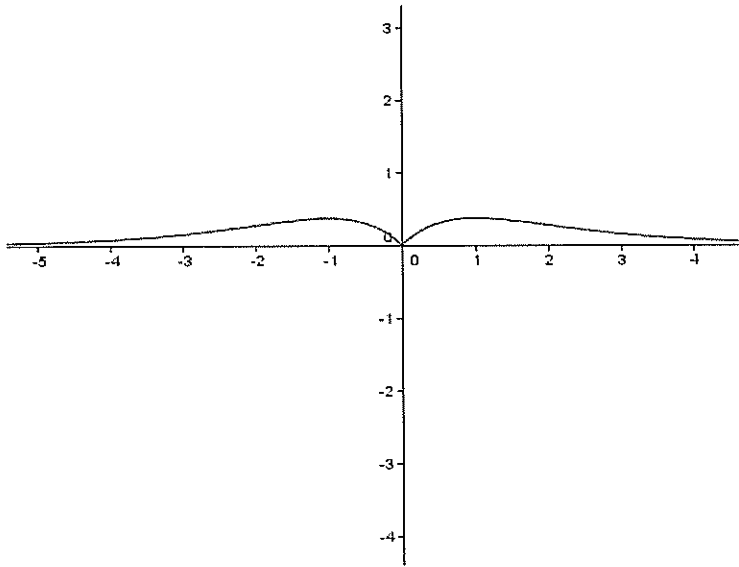
$$y = \frac{1}{f(x)}$$



$$y = x f(x)$$



$$y = f(|x|)$$



b) $9x^2 + 25y^2 = 225$

$$\frac{x^2}{25} + \frac{y^2}{9} = 1$$

(i) $b^2 = a^2(1 - e^2)$

$$9 = 25(1 - e^2)$$

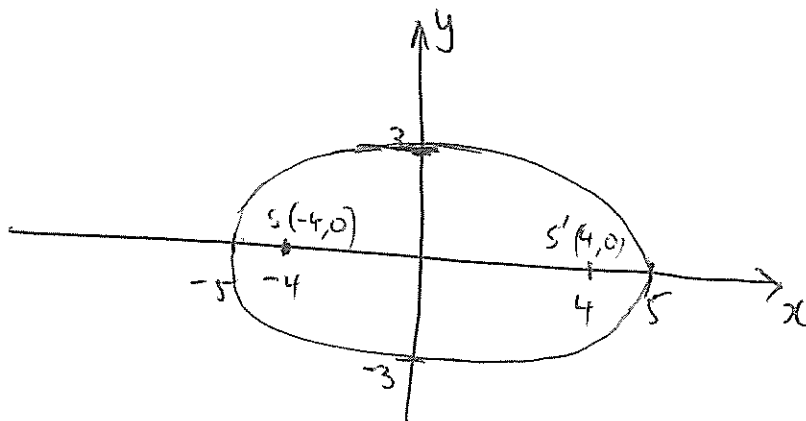
$$\frac{9}{25} = 1 - e^2$$

$$e^2 = \frac{16}{25}$$

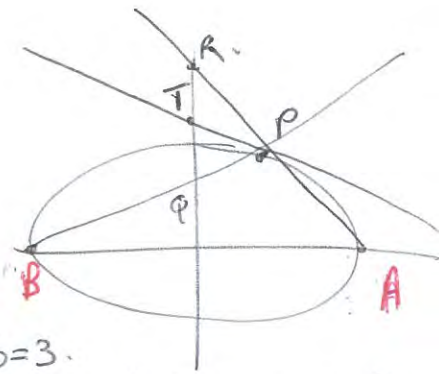
$$e = \frac{4}{5}$$

(ii) $S(ae, 0) \quad S'(-ae, 0)$

$$\therefore S(4, 0) \quad S'(-4, 0)$$



(iii)



$$a=5 \quad b=3.$$

equation of tangent at P

$$\frac{x \cos \theta}{5} + \frac{y \sin \theta}{3} = 1$$

coordinates of T $(0, \frac{3}{\sin \theta})$

AP equation of chord AP:

$$y-0 = \frac{0-3\sin \theta}{5-5\cos \theta} (x-5)$$

$$5y = -\frac{3\sin \theta}{1-\cos \theta} (x-5)$$

$$R = (0, \frac{3\sin \theta}{1-\cos \theta})$$

BP similarly equation of chord BP:

$$y-0 = \frac{0-3\sin \theta}{-5-5\cos \theta} (x+5)$$

$$Q = (0, \frac{3\sin \theta}{1+\cos \theta})$$

midpoint of Q and R

$$\begin{aligned} & \left(\frac{0+0}{2}, \frac{1}{2} \left(\frac{3\sin \theta}{1-\cos \theta} + \frac{3\sin \theta}{1+\cos \theta} \right) \right) \\ &= \left[0, \frac{3}{2} \sin \theta \left(\frac{1+\cos \theta + 1-\cos \theta}{1-\cos^2 \theta} \right) \right] \\ &= \left(0, \frac{3}{2} \sin \theta \times \frac{2}{\sin^2 \theta} \right) \end{aligned}$$

$$T = (0, \frac{3}{\sin \theta})$$

= coordinates of T.

$\therefore$ T is the midpoint of Q and R

Question 14

$$\text{a(i)} \quad \underbrace{A\left(t, \frac{1}{t}\right) \quad B\left(-t, -\frac{1}{t}\right)}_{1 : 2.}$$

$$\begin{aligned} P(x, y) &= \left(\frac{2t - t}{3}, \frac{2\left(\frac{1}{t}\right) - 1\left(\frac{1}{t}\right)}{3} \right) \\ &= \left(\frac{t}{3}, \frac{1}{3t} \right) \end{aligned}$$

$$x = \frac{t}{3}$$

$$t = 3x$$

$$\begin{aligned} y &= \frac{1}{3t} = \frac{1}{3(3x)} \\ &= \frac{1}{9x} \end{aligned}$$

$$\therefore 9xy = 1.$$

$$\text{(ii)} \quad y = \frac{1}{x}$$

$$y' = -\frac{1}{x^2} \text{ at } x = t$$

$$\text{Gradient} = -\frac{1}{t^2}$$

equation of tangent

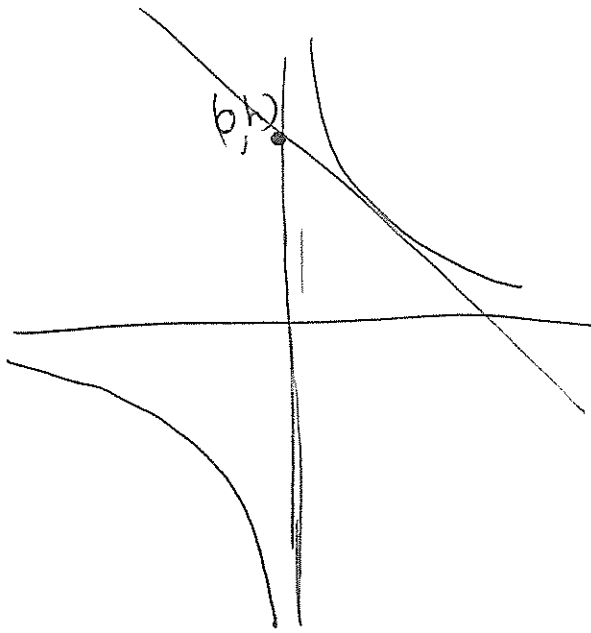
$$y - \frac{1}{t} = -\frac{1}{t^2}(x - t)$$

$$t^2 y - t = -(x - t)$$

$$t^2 y - t + x - t = 0$$

$$t^2 y - 2t + x = 0.$$

(ii)



$$t^2 y - 2t + x = 0 \text{ from (i)}$$

since it passes through $(0, h)$

$$\therefore t^2 h - 2t + 0 = 0$$

$$t(th - 2) = 0$$

$$\therefore t = \frac{2}{h} \text{ or } t = 0.$$

but $t \neq 0$.

$\therefore$ there is only one tangent
from the point $R(0, h)$

$$b) \int \frac{dx}{a^2 \cos^2 x + b^2 \sin^2 x}$$

$$= \int \frac{\sec^2 x dx}{a^2 + b^2 \tan^2 x}$$

$$\text{Let } A = \tan x \\ dA = \sec^2 x dx$$

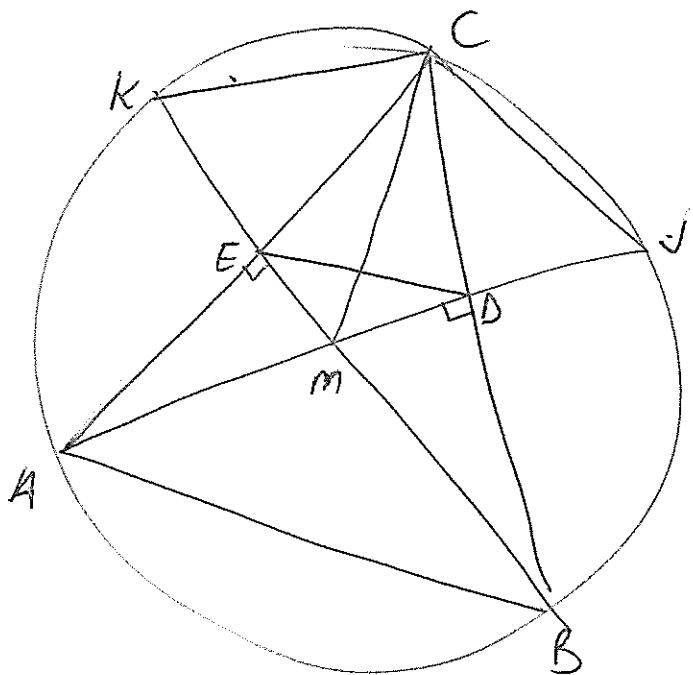
$$= \int \frac{dA}{a^2 + b^2 A^2}$$

$$= \int \frac{dA}{b^2 \left(\frac{a^2}{b^2} + A^2 \right)}$$

$$= \frac{1}{b^2} \int \frac{dA}{\left(\frac{a^2}{b^2} + A^2 \right)}$$

$$= \frac{1}{b^2} \times \frac{b}{a} \tan^{-1} \left(\frac{A}{\frac{a}{b}} \right) + C = \frac{1}{ab} \tan^{-1} \left(\frac{b}{a} \tan x \right) + C$$

C(i)



(i) $\angle CEM + \angle CDM = 180^\circ$

$\therefore$ CEMD is a cyclic quadrilateral

Since opposite sides are supplementary.

Also AB subtends 90° at D and E.

$\therefore$ ABDE is cyclic AB being the diameter.

(iii) Let $\angle KCA = \alpha$.

Since $\angle KCA = \angle KBA = \alpha$ (angles subtended by the same arc)

Similarly $\angle KBA = \angle EDA = \alpha$. (angles subtended by the same arc EM on CEMD)

$\therefore \angle KCA = \angle ACM = \alpha$

Hence AC bisects $\angle KCM$.

(iii) Join KA and BJ

In quadrilaterals $KCMA$ and $CJBM$
diagonals intersect at 90° and
bisect one pair of angles (by part ii)

$\therefore$ both quadrilaterals are kites, having
adjacent sides equal.

$$\therefore KC = MC$$

$$MC = CJ$$

$$\therefore KC = CJ,$$

hence proved.

Question 15

(a) $F = ma = -m(1+v)$

(i) $\therefore a = -(1+v)$

(ii) a) $v \frac{dv}{dx} = -(1+v)$

$$\frac{dx}{dv} = -\frac{v}{1+v}$$

$$= -\frac{1+v-1}{1+v}$$

$$= -\left(1 - \frac{1}{1+v}\right)$$

$$\frac{dx}{dv} = -1 + \frac{1}{1+v}$$

$$x = \int -1 dv + \int \frac{1}{1+v} dv$$

① — $x = -v + \ln(1+v) + C$

$x=0, t=0 \Rightarrow v=u$

② — $0 = -u + \ln(1+u) + C$

① - ② $x = -v + u + \ln\left(\frac{1+v}{1+u}\right)$
 $= u$

(b) $a = -(1+v)$

$$\frac{dv}{dt} = -(1+v)$$

$$\Rightarrow \frac{dt}{dv} = -\frac{1}{1+v}$$

① — $t = -\ln(1+v) + C$ when $t=0 \Rightarrow v=u$

② — $0 = -\ln(1+u) + C$

$$t = -\ln\left(\frac{1+v}{1+u}\right)$$

① - ②

$$\text{or } t = \ln\left(\frac{1+u}{1+v}\right)$$

$$e^t = \frac{1+u}{1+v}$$

$$\text{or } 1+v = (1+u)e^{-t}$$

$$v = (1+u)e^{-t} - 1$$

$$c) \quad \frac{dx}{dt} = (1+u)e^{-t} - 1$$

$$x = \frac{(1+u)e^{-t}}{-1} - t + C \quad t=0, x=0$$

$$0 = -(1+u) + C$$

$$\therefore C = 1+u$$

$$x = -(1+u)e^{-t} - t + (1+u)$$

$$(iii) \quad x + v + t = u.$$

$$\underbrace{-(1+u)e^{-t} - t + (1+u)}_{x} + \underbrace{(1+u)e^{-t}}_v - 1 + t$$

$$= u. \quad \text{Hence proved.}$$

$$(iv) \quad \frac{dv}{dt} = -(1+v)$$

$$\frac{dt}{dv} = -\frac{1}{1+v}$$

$$t = -[\ln(1+v)]_u^0$$

$$t = \ln(1+u) \text{ seconds}$$

$$x = -(1+u)e^{-\ln(1+u)} - \ln(1+u) + 1+u.$$

$$= u - \ln(1+u) \text{ metres}$$

$$b) P(z) = z^4 - z^3 + 6z^2 - z + 15$$

$$(i) \quad z = 1 - 2i \text{ is a factor}$$

$$\therefore \bar{z} = 1 + 2i \text{ is also a factor}$$

$$(z - (1 - 2i))(z - (1 + 2i))$$

$$= [(z - 1) + 2i][(z - 1) - 2i]$$

$$= (z - 1)^2 + 4$$

$$= z^2 - 2z + 5$$

$$\begin{array}{r} z^2 - 2z + 5 \overline{) z^4 - z^3 + 6z^2 - z + 15} \\ \underline{z^4 - 2z^3 + 5z^2} \\ z^3 + z^2 - z \\ \underline{z^3 - 2z^2 + 5z} \\ 3z^2 - 6z + 15 \\ \underline{3z^2 - 6z + 15} \\ 0 \end{array}$$

$$P(z) = (z^2 - 2z + 5)(z^2 + z + 3)$$

$$(ii) \quad \text{Solve for } P(z) = 0$$

$$z^2 - 2z + 5 = 0, \quad z^2 + z + 3 = 0$$

$$z = 1 \pm 2i$$

$$z = \frac{-1 \pm \sqrt{11}i}{2}$$

Question 16

$$a_{(i)} \quad \frac{1}{2^{p+1}} + \frac{1}{2^{p+2}} > \frac{1}{p+1}$$

$$\text{now } 2^{p+1} < 2^{p+2}, \quad p > 0$$

$$\therefore \frac{1}{2^{p+1}} > \frac{1}{2^{p+2}} \quad \left(\text{if } a < b \Rightarrow \frac{1}{a} > \frac{1}{b} \right)$$

$$\therefore \frac{1}{2^{p+1}} + \frac{1}{2^{p+2}} > \frac{1}{2^{p+2}} + \frac{1}{2^{p+2}}$$

$$= \frac{2}{2(p+1)}$$

$$= \frac{1}{p+1}$$

$$\therefore \frac{1}{2^{p+1}} + \frac{1}{2^{p+2}} > \frac{1}{p+1}$$

$$(ii) \quad Y(m) = \frac{1}{m+1} + \frac{1}{m+2} + \dots + \frac{1}{2m} \geq \frac{37}{60}$$

Step 1 Test for $m=3$.

$$Y(3) = \frac{1}{4} + \frac{1}{5} + \frac{1}{6} = \frac{37}{60} \quad \text{R.N.S.}$$

Also for $m=4$

$$Y(4) = \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} = \frac{37}{60} + \frac{1}{7} > \frac{37}{60}$$

so the results are true for $m=3$ and $m=4$

Step 2 Let the result be true for $m=k \quad k \geq 3$

$$\text{i.e. } Y(k) = \frac{1}{k+1} + \frac{1}{k+2} + \dots + \frac{1}{2k} \geq \frac{37}{60}$$

show the result is true for $m = k+1$

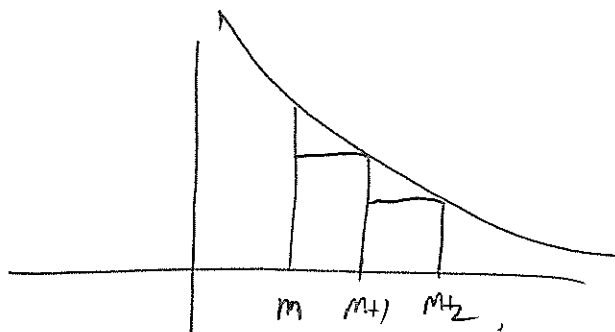
$$\text{ie } \frac{1}{k+2} + \frac{1}{k+3} + \dots + \frac{1}{2k+2} \geq \frac{37}{60}$$

L.H.S.

$$\begin{aligned} & \frac{1}{k+2} + \frac{1}{k+3} + \dots + \frac{1}{2k} + \frac{1}{2k+1} + \frac{1}{2k+2} \\ &= \underbrace{\frac{1}{k+1} + \frac{1}{k+2} + \frac{1}{k+3} + \dots + \frac{1}{2k} + \frac{1}{2k+1} + \frac{1}{2k+2}}_{\geq \frac{37}{60}} - \frac{1}{k+1} \\ &\geq \frac{37}{60} + \left(\frac{1}{2k+1} + \frac{1}{2k+2} - \frac{1}{k+1} \right) \\ &\geq \frac{37}{60} \text{ as } \frac{1}{2k+1} + \frac{1}{2k+2} > \frac{1}{k+1} \text{ using part a.} \end{aligned}$$

$\therefore$ proved
 $\therefore$ true by mathematical induction.

(iii)



$$\text{low rect} < \int_m^{m+1} \frac{1}{t} dt < \text{upper rectangle}$$

$$\text{Area of lower rectangle} = 1 \left(\frac{1}{m+1} \right)$$

$$\therefore \int_m^{m+1} \frac{1}{t} dt > \frac{1}{m+1}$$

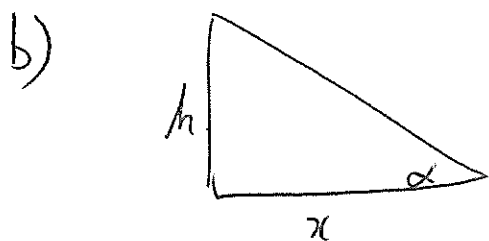
$$(v) \int_m^{2m} \frac{1}{t} dt = \int_m^{m+1} \frac{1}{t} dt + \int_{m+1}^{m+2} \frac{1}{t} dt + \dots + \int_{2m-1}^{2m} \frac{1}{t} dt$$

$$> \frac{1}{m+1} + \frac{1}{m+2} + \dots + \frac{1}{2m} \quad (\text{using (iv)})$$

$$> \frac{37}{60} \quad (\text{using part ii})$$

$$\begin{aligned} \text{Also } \int_m^{2m} \frac{1}{t} dt &= [\ln t]_m^{2m} \\ &= \ln 2m - \ln m \\ &= \ln 2. \end{aligned}$$

$$\therefore \ln 2 > \frac{37}{60}.$$



$$\begin{aligned} \text{(i) Area} &= \frac{1}{2} h \times x \\ &= \frac{1}{2} x \tan \alpha \times x \\ &= \frac{1}{2} x^2 \tan \alpha \end{aligned}$$

$$\begin{aligned} x^2 + y^2 &= r^2 \\ x^2 &= r^2 - y^2 \end{aligned}$$

$$\therefore A = \frac{1}{2} (r^2 - y^2) \tan \alpha.$$

$$(ii) \quad \int V = \frac{1}{2} (r^2 - y^2) \tan \alpha \, dy$$

$$V = \lim_{r \rightarrow 0} \sum_{-r}^r \frac{1}{2} (r^2 - y^2) \tan \alpha \, dy$$

$$= \frac{1}{2} \tan \alpha \int_{-r}^r (r^2 - y^2) dy$$

$$= \frac{2}{2} \tan \alpha \int_0^r (r^2 - y^2) dy \quad (\text{since } r^2 - y^2 \text{ is an even function})$$

$$= \tan \alpha \left[r^2 y - \frac{y^3}{3} \right]_0^r$$

$$= \tan \alpha \left[r^3 - \frac{r^3}{3} \right]$$

$$= \frac{2}{3} r^3 \tan \alpha \quad \text{cubic units}$$